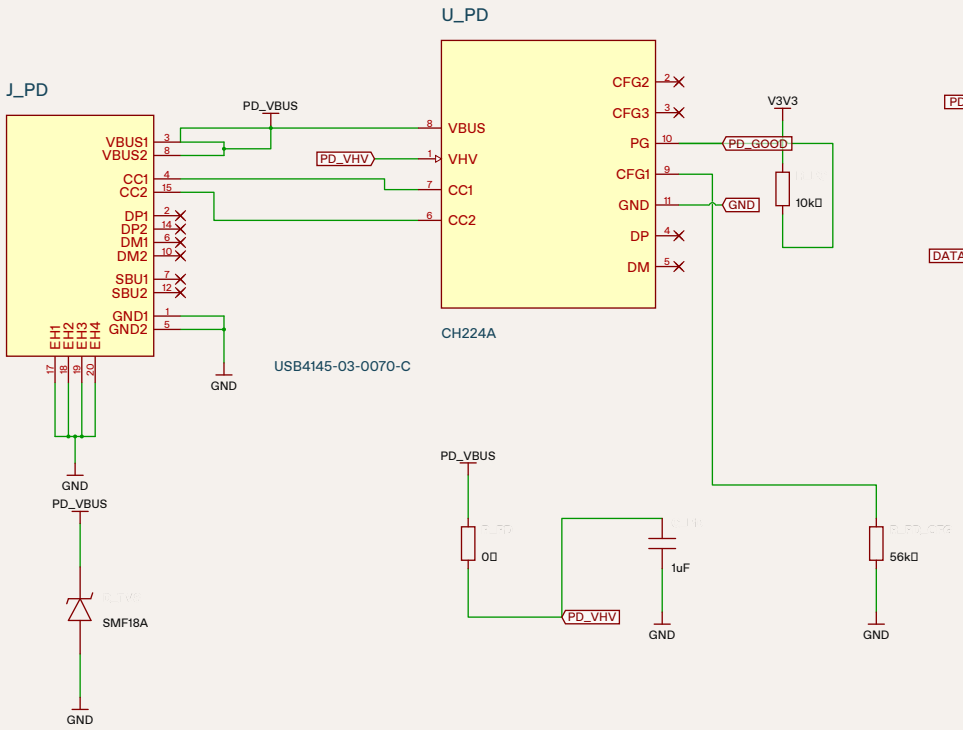
# NEMA11 / RP2040

## Power & USB-PD

Motor supply negotiation, reverse-blocked logic power and 3.3 V conversion

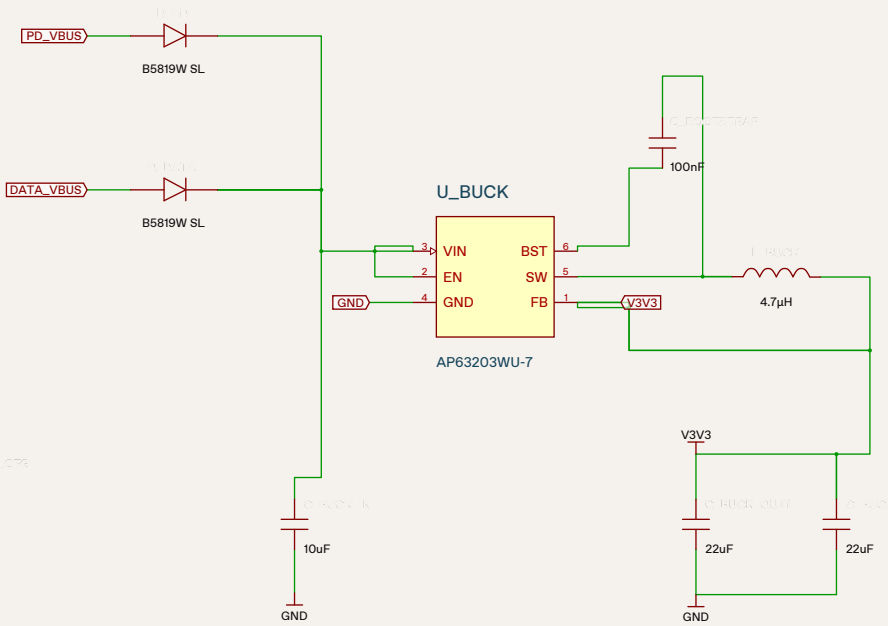
### USB-C motor power

Fixed 15 V request; data and SBU contacts unused




### 3.3 V logic supply

AP63203 fixed-output buck; USB power is diode-isolated




CH224A: 56 kΩ CFG1-to-GND selects 15 V; CFG2/3 NC.

C_PD: 1 µF / 50 V (C15849); R_PD: 0 Ω (C21189).

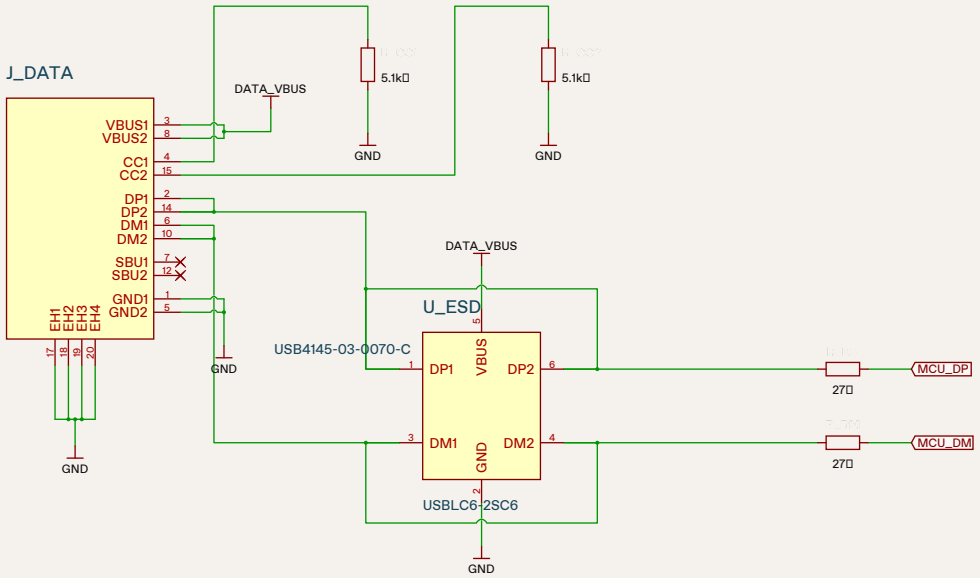
PD_VBUS = motor rail DATA_VBUS = programming USB 5 V

VIN range and transients require bench qualification.

Programming USB-C, data-line protection and nonvolatile boot interfaces

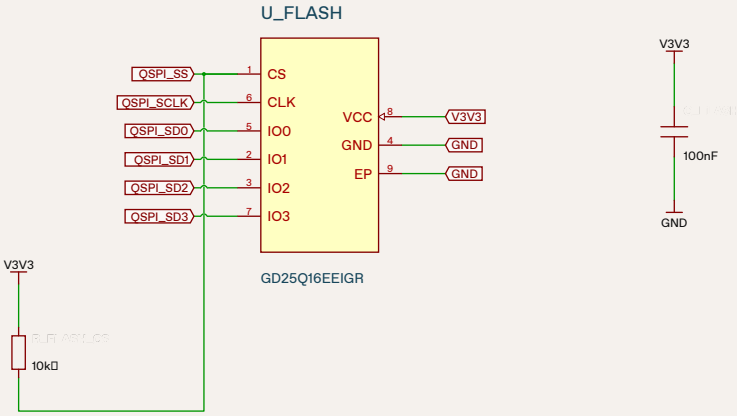
## USB-C data port

Separate 5.1 kΩ CC pull-downs; duplicated D+/D− contacts



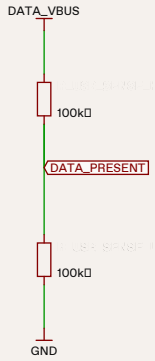

## QSPI flash & boot selection

GD25Q16E; BOOT contact asserts the boot input




## USB presence

Resistive divider to MCU; fixed 5 V input only

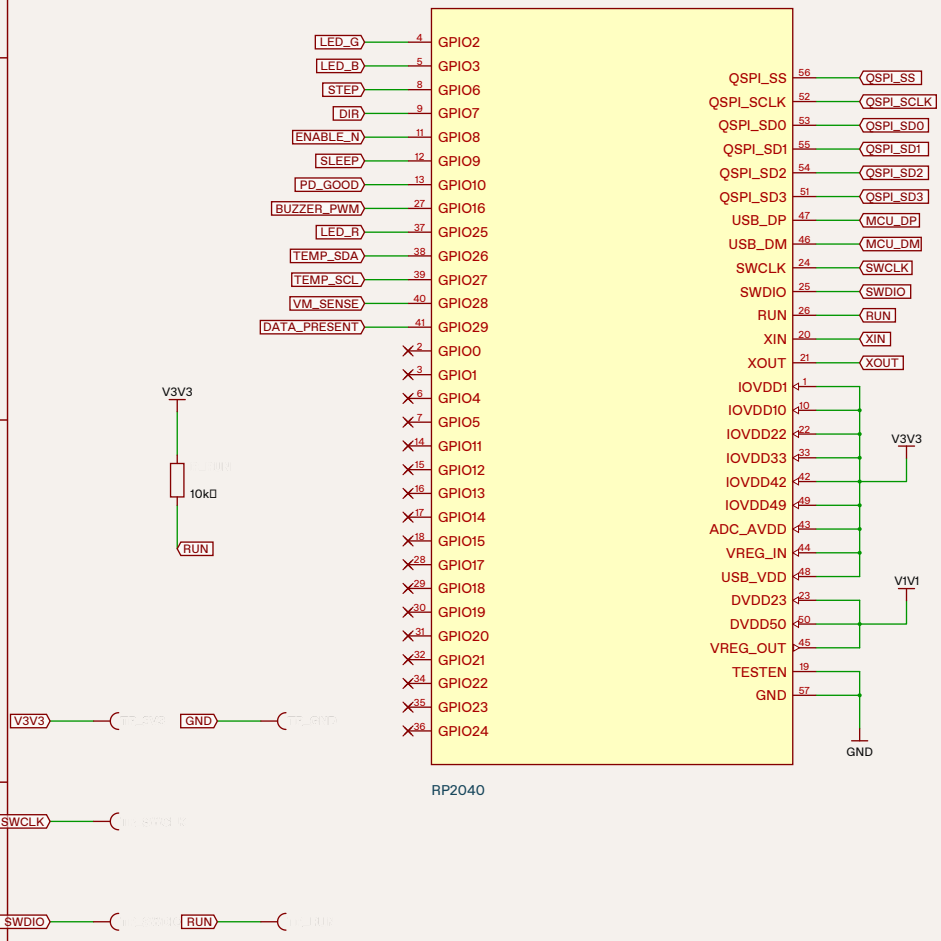





# Controller & clocks

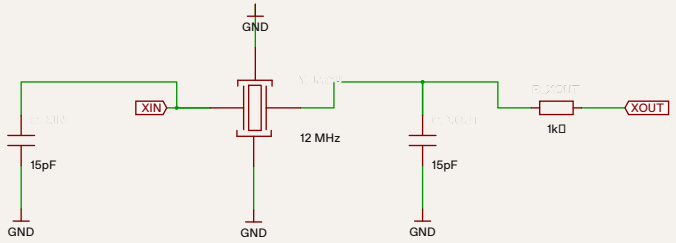
MCU interfaces, reset/debug access, power decoupling and clock support

U_MCU



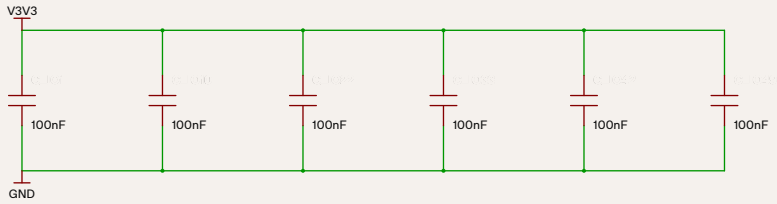

## 12 MHz crystal

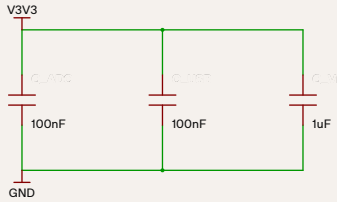
Native 4-pin crystal; load capacitors and series resistor




## 3.3 V bypass

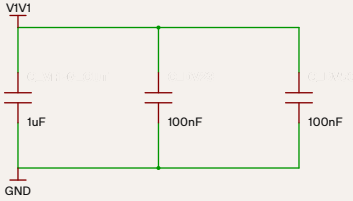
IOVDD, ADC and USB decoupling




## 1.1 V core

On-chip VREG_OUT feeds DVDD



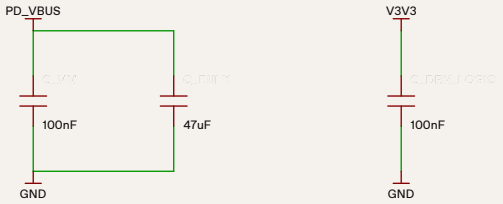
# NEMA11 / RP2040

## Stepper motor driver

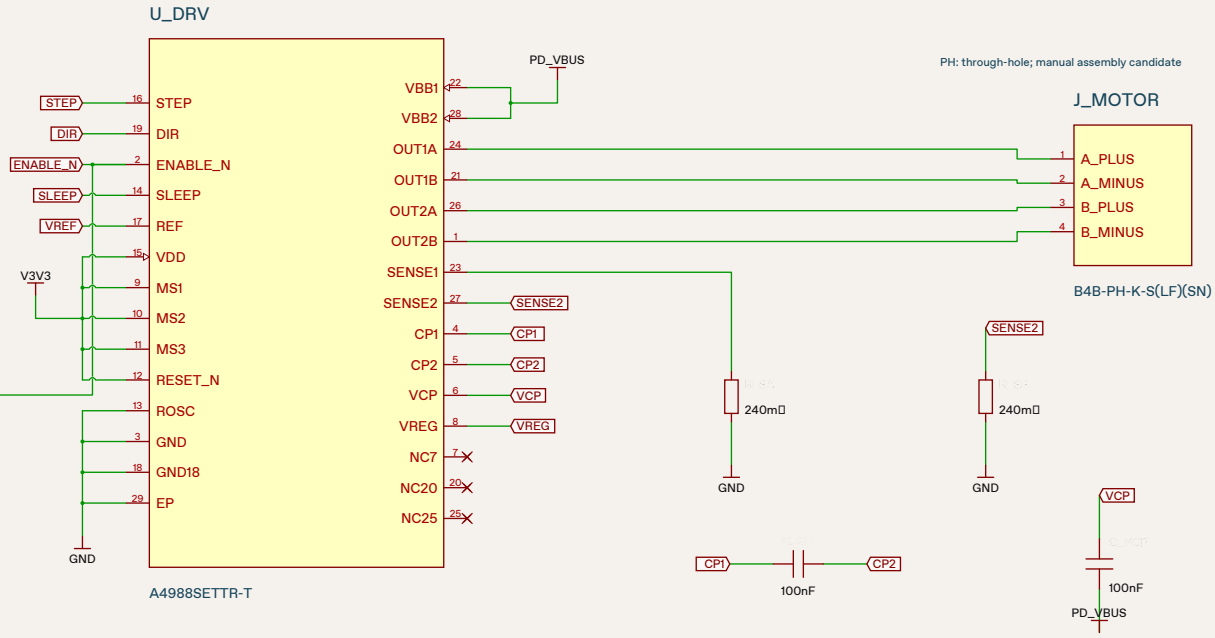
A4988 phase outputs, charge pump, current limit and motor-voltage measurement

### Driver supplies

Local bulk and high-frequency bypass

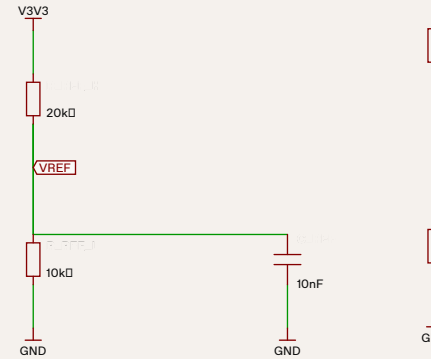


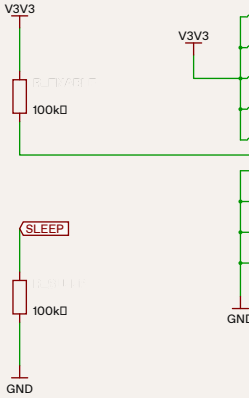

JST-PH pin order: 1 A+ 2 A– 3 B+ 4 B–




### Current reference

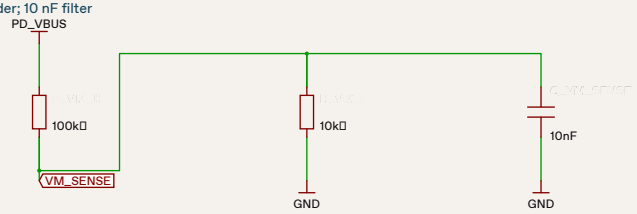
VREF / (8 × RSENSE) = 0.573 A nominal; verify waveform and thermal limits




### Motor voltage monitor

100 kΩ / 10 kΩ divider; 10 nF filter



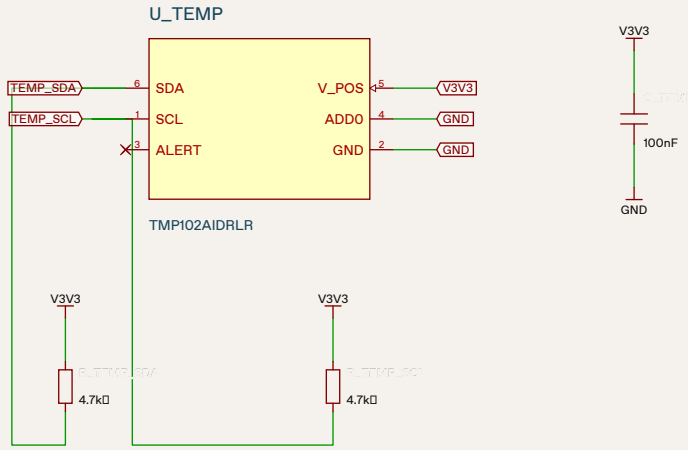
# NEMA11 / RP2040

## Temperature & feedback

I²C temperature telemetry, RGB status outputs and PWM magnetic buzzer

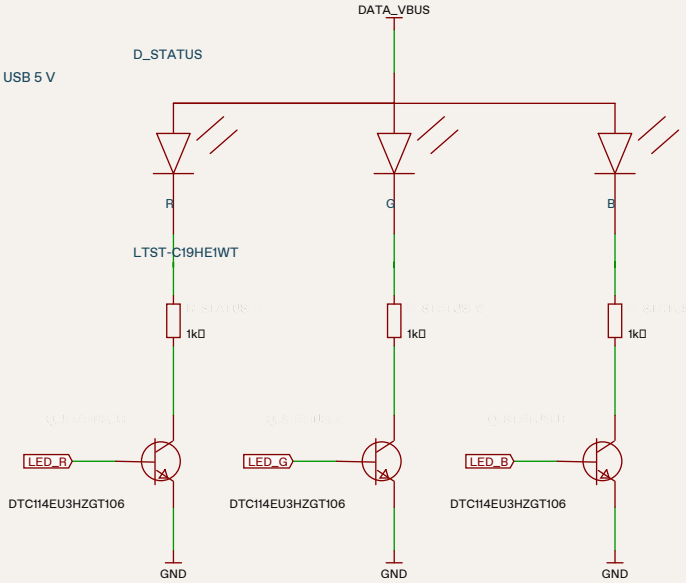
### TMP102 temperature

ADD0 grounded: address 0x48; ALERT unused




### RGB status indicator

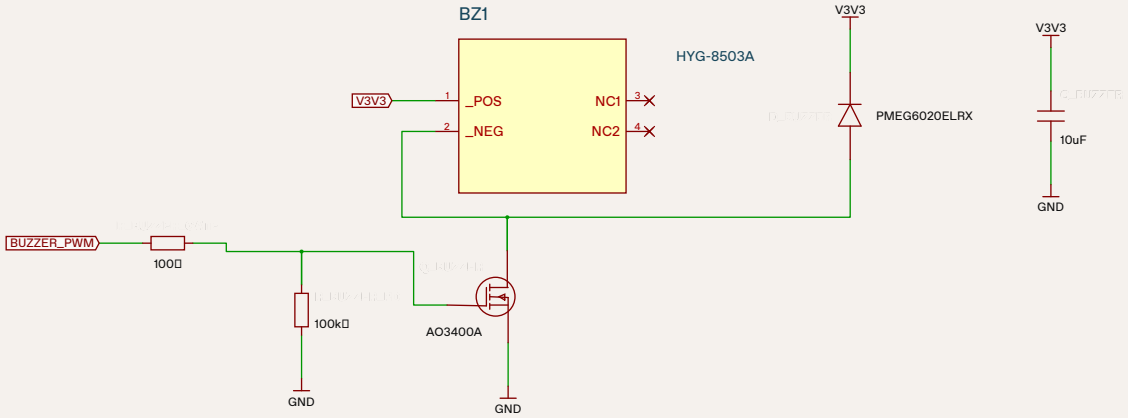
Common anode uses programming USB 5 V




DTC114E: NPN with internal input/bias resistors.

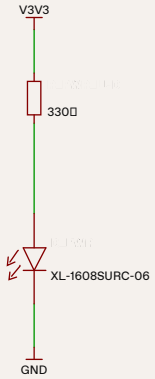
### PWM buzzer

120 mA magnetic load; MOSFET drive and flyback diode




### Power indication

3.3 V rail present; RGB/buzzer require firmware




AnasSarkiz / RP2040 motor controller

35 × 40 mm | 4-layer PCB | Hardware prototype — assembly/firmware/bench qualification pending

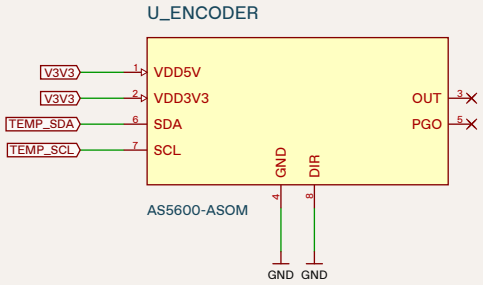
# NEMA11 / RP2040

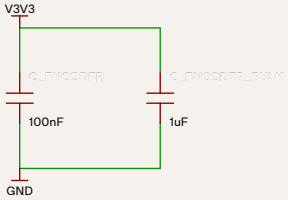
## Optional encoder (DNP)

Bottom footprints only; AS5600 and both bypass capacitors omitted from factory assembly

### Future magnetic shaft position

Nominal sensor centre (0, 0); rear shaft/magnet geometry remains unverified




U_ENCODER, C_ENCODER and C_ENCODER_BULK: bottom / DO NOT PLACE.

3.3 V mode: VDD5V + VDD3V3 tied together; DIR grounded; OUT/PGO NC.

Shared existing temperature I²C bus: AS5600 0x36 / TMP102 0x48.

Existing 4.7 k□ top-side pull-ups retained. Encoder firmware is required.

A diametrically magnetized shaft magnet and motor fit must be validated.